MAT367
Intro to Differential Geometry

Class Lecture Notes

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January 6, 2026

Last Updated:

March 12, 2026

Primary source material:

Class Lectures

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Preface

These notes were created during class lectures. As such, they may be incomplete or lacking in some detail at parts, and may contain confusing typos due to time-sensitivity. Additionally, these notes may not be comprehensive. Most statements in this document which are not Theorems, Problems, Lemmas, Corollaries, or similar, are likely paraphrased to a certain degree. Please do not treat any material in this document as the exact words of the original lecturer.

If you are viewing this document in Obsidian, you may notice that the links in the pdf document do not work. This is intentional behaviour, as I currently do not have or know of a decent solution which allows them to behave well with the setup in Obsidian. However, below certain pages, there may be links to other documents - these are usually context-relevant links between notes of different areas of study. I created these links to point out potential similarities, or in case one area of study is borrowing a concept, definition, or theorem from another area of study, and you wish to see the full, original definition/derivation/proof or whatever it may be.

V Vector Fields and Bundles

8 Vector Fields

(check) Lec 13 - Feb 24 (Week 7)

we jump arnd a bit - doing chs 6/9 kinda together.

whats a vector field? morally, a smooth choice of tangent vec at every pt in M. eg, for $M \subseteq \mathbb{R}^n$, tangent vecs live in $T_pM \subseteq \mathbb{R}^n$.

Definition 8.1

a **vector field** is a smooth map $X : M \longrightarrow \mathbb{R}^n$ s.t. $X(p) \in T_pM \subseteq \mathbb{R}^n$.

what does it mean on an abstract mfld? "lets start by cheating". v1: look at charts.

Definition 8.2

a **vector field** X on M is a map $p \mapsto X(p) \in T_pM$, such that for each chart φ, we have $D_p\varphi \circ X : U \longrightarrow \mathbb{R}^n$ is smooth.

this uses the identification $T_pM \simeq \mathbb{R}^n$ induced by chart. but what is that thing between $D_p\varphi \circ X$?

Definition 8.3

the **tangent bundle** TM of M^m is a smooth $2m$-mfld with:

- pt set $\bigsqcup_{p \in M} T_pM$
- (non-max'l) atlas w charts of the form:

$$\left(\bigsqcup_{p \in U} T_pM, (\varphi \circ \pi, D\varphi) \right) \qquad (\varphi \circ \pi, D_p\varphi) : \bigsqcup_{p \in U} T_pM \longrightarrow \mathbb{R}^m \times \mathbb{R}^m$$

for charts (U, φ) in M.

check: π is smooth. remarks:

(1) the tangent bundle comes w a proj $\pi : TM \longrightarrow M$, given by $T_pM \mapsto p$.

(2) for each $p \in M$, the preimg (or fiber) $\pi^{-1}(p)$ is a vspace.

(3) *local triviality:* for each $p \in M$, there's a nbhd $U \ni p$ s.t. the diagram commutes: [TODO: add diagram], where $(D\varphi, \pi)$ is a smooth map and linear on each $\pi^{-1}(q)$.

Lec 15 - Mar 5 (Week 8)

accidentally git yeeted tuesdays notes so those are gone forever

last time: a VF X on M defns a flow $\Phi : J \subseteq \mathbb{R} \times M \longrightarrow M$ where $\dot{\Phi}_t(p) = X_{\Phi_t(p)}$, for some $J \ni \{0\} \times M$ open.

Definition 8.4

a VF is **complete** if $J = \mathbb{R} \times M$.

for instance, $x\partial_x$ is complete w $\Phi_t(x) = e^{tx}$, but $x^2\partial_x$ is not (escapes to ∞ in finite time).

Theorem 8.5

if X has cpt sppt, it is complete.

Proof.

1) if $p \notin \operatorname{supp}(X)$, $X_p = 0$ implies $\Phi_t(p) = p$ for all t. if $p \in \operatorname{supp}(X)$, then $\Phi_t(p) \in \operatorname{supp}(X)$ when defnd.

2) since J open, $\forall \varepsilon > 0$, there is an open $(-\varepsilon, \varepsilon) \times U_\varepsilon$, where:

$$U_\varepsilon \;=\; \{p \in M : \Phi_t(p) \text{ defnd for } t \in (-\varepsilon, \varepsilon)\}$$

these cvr M. by cptness, take finite subcvr $\{U_{\varepsilon_i}\}$. then $U_{\varepsilon_{\min}}$=M.

3) by flow property, $\Phi_{\varepsilon_{\min}/2}$ defnd $\forall p \in M$. thus $\Phi_{n\varepsilon_{\min}/2}(p)$ is defined on $t \geq 0$, and $h < 0$ can be handled the same way.

■

summary: if X a complete VF, its flow defns a 1-param. family of diffeos.

Corollary 8.6

any VF on a cpt mfld is complete.

is the crll out of order? oh well. note:

i) $\Phi_t \circ \Phi_s = \Phi_{t+s}$

ii) $\Phi(t, p)$ is a smooth fn on $\mathbb{R} \times M$

so $\Phi(t, p)$ a smooth fn on $\mathbb{R} \times M$.

Proposition 8.7

if Φ_t satisfies i) and ii) above, its the flow of a VF X, where:

$$X_p(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\Phi_t(p))$$

in other words, X_p is the vel vec of $\gamma(t) = \Phi_t(p)$.

Proof.

first, X satisfies the product rule:

$$X_p(fg) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(\Phi_t(p))g(\Phi_t(p))$$

its a VF. trust. it is also an integral curve of X, as:

$$\dot{\Phi}_t(p) = \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \Phi_s(\Phi_t(p)) = X_{\Phi_t(p)}$$

sure. ■

example: given $A \in M_n(\mathbb{R})$, let $\Phi_t : \mathbb{R}^m \longrightarrow \mathbb{R}^m$ be:

$$\Phi_t(x) \;=\; e^{tA}x \;=\; \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} A^n\right) x$$

then the corresp. VF is:

$$X(f)(x) \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(e^{tA}x) \;=\; \nabla f \cdot \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} e^{tA}x \;=\; \nabla f \cdot Ax$$

so in "wishy-washy physics notation", we have $X = Ax \cdot \nabla$.

now, an alt defn of the lie bracket. we start with, naturally, the pullback.

Definition 8.8

given $F : M \longrightarrow N$, we denote by $F^* f = f \circ F$ the **pullback** of $f : N \longrightarrow \mathbb{R}$.

if F is a diffeo, then for a VF X on N, we can define $F^* X$ by:

$$F^* X_{F^{-1}(p)} = TF^{-1}(X_p)$$

Definition 8.9

the **lie derivative** of $f : M \longrightarrow \mathbb{R}$ wrt a VF X is given as:

$$L_X f \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^* f \;\in\; C^\infty(M)$$

the lie derivative of a VF Y wrt X is:

$$L_X Y \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^* Y \quad \in \mathfrak{X}(M)$$

where Φ is the flow of X.

check: $L_X f = X(f)$.

Theorem 8.10

$L_X Y \;=\; [X, Y]$.

Lec 16 - Mar 10 (Week 9)

recall the defn of the lie deriv. of a VF. since Φ is a diffeo, it induces an isomorphism $T_pM \xrightarrow{T_p(\Phi_t)} T_{\Phi_t(p)M}$. this gives a way to compare vectors in these spaces, as $\Phi_t^* Y$ pulls tangent vectors back into T_pM (i.e. all vecs $\Phi_t^* Y \in T_pM$).

Proof.

[of 8.10] given $f \in C^\infty(M)$, wts $L_X Y(f) = [X, Y](f)$. we have:

$$\Phi_t^*(Y(f)) \;=\; (\Phi_t^* Y)(\Phi_t^* f)$$

taking deriv at $t = 0$, we get:

$$X(Y(f)) \;=\; \left(\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^*(Y)\right)(f) + Y(\Phi_t^* f) \;=\; L_X Y(f) + Y(X(f))$$

thus $L_X Y = XY - YX = [X, Y]$ as needed. ∎

remark: $L_X f := \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \Phi_t^* f$ measures the change in f along an integral curve(?) of X, that is, a curve whose velocity vector at $t = 0$ is X_p. this is precisely $X(f)$. thus, $[X, Y]$ is how Y changes along the flow of X. in particular, we have the following:

Theorem 8.11

sps Φ flow of X, Ψ flow of Y. then:

$$[X, Y] = 0 \iff \Phi_t^* Y = Y \iff \Psi_t^* X = X \iff \Phi_t \circ \Psi_s = \Psi_s \circ \Phi_t$$

for all s, t.

Proof.

the first two equivalences are clear from the definition. if $\Phi_t^*(Y) = Y$, then Φ_t takes integral curves of Y to integral curves of Y. uhm.

he literally just gave up lmao. ■

more generally, we have that:

$$[X, Y] \;=\; \lim_{t \to 0} \frac{1}{t^2}[\Phi_t^* \Psi_t^* - \Psi_t^* \Phi_t^*]$$

lets not attempt to prove it this time.

and now, an interpretation of the jacobi identity. recall:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] \;=\; 0$$

we can rewrite this into the following form (which saany uses):

$$[X, [Y, Z]] \;=\; [[X, Y], Z] + [Y, [X, Z]]$$

this can be interpreted as a product rule, where $[\cdot, \cdot]$ is both the derivation and the product. "the lie bracket on derivations is a derivation on lie brackets".

a theorem! when can you take lin. indep. VF $X_1, \ldots, X_r$ and "integrate them" to get an r-dim submfld S? i.e., for every $p \in M$, there is locally a unique r-dim submfld S thru p s.t. $X_1, \ldots, X_r$ are tangent to S near p.

Theorem 8.12: Frobenius' Theorem

the above happens iff:

$$[X_i, X_j] \;=\; \sum_{k=1}^{r} c_{ij}^k(p) X_k$$

for all i, j. this is called the *frobenius condition.*

as an example, consider $M = T^2$ and $X = 2\partial_x + 3\partial_y$. representing T^2 as a quotient of the square, the flow line of X can be thought of as a line with slope $3/2$. all of the flow

lines is, intuitively, called a *foliation by submflds.* in particular, the flow lines would "be" a submanifold.

if on the other hand, we consider $X = \partial_x + \sqrt{2}\partial_y$, a single flow line would densely cover T^2. the flow lines also define a foliation, i.e. a locally nice-looking partition into submflds, but globally they are not submflds.

Lemma 8.13: Flow-straightening lemma

given lin. indep. $X_1, \ldots, X_r$ VF on M w $[X_i, X_j] = 0$ and p s.t. $X_p \neq 0$, $\exists (U, \varphi)$ near p s.t. $X_i = \partial_i$ wrt the chart.

Proof.

choose a codim r N s.t. $\operatorname{span}((X_i)_p, T_pN) = T_pM$. consider $F : (-\varepsilon, \varepsilon)^r \times N \longrightarrow M$ given as:

$$F(t_1, \ldots, t_r, p) \;=\; \Phi_{t_1}^{x_1} \circ \cdots \circ \Phi_{t_r}^{x_r}(p)$$

note the flows can be rearranged, and we have $TF(\partial_{t_i}) = X_i$. then $T_{(0,p)}F(s, v) = v + \sum s_i (X_i)_p$, so Φ full rank at $(0, p)$. by inv fn thm, we get a chart. ■

pf of frebenius thm: forward dir is true since if X_i, X_j are tangent to submfld S, then so is $[X_i, X_j]$. thus, its in their span. OTOH, we reduce to $[\cdot, \cdot] = 0$ case. we have (?) in $U \subseteq \mathbb{R}^m$ in coords:

$$X_i = \sum_{j=1}^{r} a_{ij}(x)\partial_j + \sum_{j=r+1}^{m} b_{ij}(x)\partial_j$$

by a lin. change of coords, $a_{ij}(0) = I$ and $b_{ij}(0) = 0$. define $X_i = \sum_{j=1}^{r} a_{ij} X'_j$. the X'_j span the same thing and commute, so by the lemma, they give a submfld.

Lec 17 - Mar 12 (Week 9)

TODO: sort these notes. ... are we reviewing?

to understand VF, need to understand them as:

- *coord-wise*: $X = \sum X_i \partial_i$, where X_i's are smooth fns (in any chart on M)

- *abstract*: a smooth section of the tangent bundle (smooth choice of tangent vec for each point)
- *functional* [or sheaf-theoretic?]: a derivation on $C^\infty(M)$
- *integral*: sth that generates a flow (a 1-param family of diffeos)

we can do sth similar for diff forms.

- *coord-wise*: k-form $\omega = \sum f_I \, \mathrm{d}\omega_{i_1} \wedge \cdots \wedge \mathrm{d}\omega_{i_k}$
- *abstract*: ω is a smooth choice of k-covec at each point (a section of some [the cotangent] vector bundle)
- *functional*: ω a kind of fn $(\mathfrak{X}(M))^k \longrightarrow C^\infty(M)$
- *integral*: a k-form is sth integrable over a k-mfld

Example 8.14

given $f \in C^\infty(M)$, defn its (exterior) differential $\mathrm{d}f : \mathfrak{X}(M) \longrightarrow C^\infty(M)$ by:

$$\mathrm{d}f(X) = X(f)$$

note, $\mathrm{d}f$ is $C^\infty(M)$-linear, i.e.:

$$\mathrm{d}f(gX + hY) = g\mathrm{d}f(X) + h\mathrm{d}f(Y)$$

that is an exercise.

Definition 8.15: Functional

a **1-form** on M is a $C^\infty(M)$-linear map $\mathfrak{X}(M) \longrightarrow C^\infty(M)$. the set ($C^\infty(M)$-alg) of 1-forms is denoted $\Omega^1(M)$.

hey, rmbr dual vspaces? given $\alpha \in V^*$, we can write $\alpha(v)$ as $\langle \alpha, v \rangle$. this emphasizes that if V finite-dim, $(V^*)^* = V$ canonically. that is, we can think of V as a hom $V^* \longrightarrow \mathbb{R}$. note $V^* \simeq V$, but not canonically unless we first fix an inner prod.

Definition 8.16

given basis $e_1, \ldots, e_n$ of V, the **dual basis** for V^* is $e^1, \ldots, e^n$, where $e^i(e_j) = \delta_{ij}$.

cotangent space is defnd precisely as u think. its elems are called **covectors**. in $T_p\mathbb{R}^n$, we have a distinguished basis $\partial_1, \ldots, \partial_n$. its dual basis is written $\mathrm{d}x_1, \ldots, \mathrm{d}x_n$, where:

$$\mathrm{d}x_i(a_1\partial_1 + \cdots + a_n\partial_n) = a_i$$

Definition 8.17

given a lin. map $F : V \longrightarrow W$, theres a **dual map** $F^* : W^* \longrightarrow V^*$ defn'd by $F^*\omega(v) = \omega(Fv)$.

given smooth map $f : M \longrightarrow N$, we have a linear map $T_pf : T_pM \longrightarrow T_{f(p)}N$ given by $v \mapsto f_*v = T_pf(v)$. therefore, $T_p^*f : T_p^*M \longrightarrow T_{f(p)}^*N$ is given as $v \mapsto f^*v = T_p^*f(v)$. the "base" map is as a pushforward, i.e. vecs are *covariant*. the dual is as a pullback, i.e. covecs are *contravariant*.

Example 8.18

given $f \in C^\infty(M)$, we can defn a covec $\mathrm{d}_pf \in T_p^*M$ by:

$$\mathrm{d}_pf(v) = v(f) = D_vf$$

in particular, in $\mathbb{R}^n$, we can write $\mathrm{d}_pf = \partial_{x_1}f\mathrm{d}x_1 + \cdots + \partial_{x_n}f\mathrm{d}x_n$. exercise... lol.

Proposition 8.19

given $f \in C^\infty(M, N), g \in C^\infty(N)$, we have $\mathrm{d}_p(f^*g) = f^*\mathrm{d}_pg$.

Proof.

for any $v \in T_pM$:

$$\begin{aligned}
\langle \mathrm{d}_p(f^*g), v\rangle &= v(f^*g) \\
&= v(g \circ f) \\
&= (T_pf(v))(g) \\
&= \left\langle \mathrm{d}_{f(p)}g, T_pf(v)\right\rangle \\
&= \left\langle f^*\mathrm{d}_{f(p)}g, v\right\rangle
\end{aligned}$$

note $\langle \cdot, \cdot \rangle$ is not an inner prod here. ■